0021. Merge Two Sorted Lists

sy | Category: Linked List | LeetCode: https://leetcode.com/problems/merge-two-

Problem

- Merge two sorted linked lists into one sorted list by splicing their nodes.
- Example: 1→2→4 and 1→3→4 → 1→1→2→3→4→4.

Intuition

- Like the merge step of merge sort: repeatedly take the smaller head.
- A dummy node removes the special case for the first node.

Brute Force

- Collect all values, sort, and rebuild: O((m+n) log(m+n)).

Optimal Approach

- Dummy → tail.
- While both non-null: attach the smaller node, advance that list and tail.
- Attach the remaining list.

Worked Example

```
a = 1->2->4, b = 1->3->4.

1 vs 1 -> take a1   | 2 vs 1 -> take b1
2 vs 3 -> take a2   | 4 vs 3 -> take b3
4 vs 4 -> take a4   | a empty -> append b4
result 1,1,2,3,4,4
```

Java Solution

```java
public class MergeTwoSortedLists {
    public ListNode mergeTwoLists(ListNode a, ListNode b) {
        ListNode dummy = new ListNode(0), tail = dummy;
        while (a != null && b != null) {
            if (a.val <= b.val) { tail.next = a; a = a.next; }
            else { tail.next = b; b = b.next; }
            tail = tail.next;
        }
        tail.next = a != null ? a : b;
        return dummy.next;
    }
}
```

Complexity

- Time: O(m+n). Space: O(1).

Pitfalls

- Forgetting to append the leftover list.